



Rizwan Abid

Nationality: Pakistani **Date of birth:** 01/01/1987 **Gender:** Male

Phone number: (+92) 3224069330 **Phone number:** (+92) 3224069330

Email address: rizwan-abid@live.com

LinkedIn: [linkedin.com/in/rizwan-abid-phd-a9362b82](https://www.linkedin.com/in/rizwan-abid-phd-a9362b82)

Website: <https://www.researchgate.net/profile/Rizwan-Abid>

Home: 54840 Lahore (Pakistan)

ABOUT ME

Dr Rizwan Abid is a genomics and molecular biology professional with more than six years of specialized experience in genomics, molecular biology, sequencing technologies, and genomic data analysis. He brings a strong combination of scientific expertise, technical proficiency, and research leadership.

His core expertise encompasses NGS, genomic data analysis, and high throughput sequencing workflows, with wide experience across Illumina platforms, including MiniSeq, MiSeq i100, NextSeq, and NovaSeq.

His research portfolio spans cancer genomics, neurological disorders, brain tumors, with a focus on the identification and validation of genetic markers through advanced sequencing and molecular biology methodologies.

Dr Rizwan also works at the intersection of artificial intelligence, machine learning, and genomics, with a particular interest in predictive analytics and precision healthcare. His approach combines scientific rigor with translational thinking to derive meaningful biological and clinical insights from complex genomic data.

In addition to his scientific and technical capabilities, he has demonstrated strengths in project coordination, scientific communication, problem solving, and multidisciplinary collaboration. He is committed to leveraging advanced genomic technologies and emerging computational approaches to strengthen research outcomes and contribute to high impact solutions across healthcare and life sciences.

WORK EXPERIENCE

Visole Group, GMS, Illumina | Next Generation Sequencing group – Lahore, Pakistan

Genomics and Sequencing Specialist

[12/08/2024 – Current]

- Provide expert support on Illumina NGS platforms (iSeq, MiSeq, HiSeq, NovaSeq)
- Optimizing protocols and workflows for customer success.
- Deliver training on NGS instruments and data analysis tools
- Troubleshoot technical issues and provide ongoing post-sales support
- Collaborate with customers on research initiatives.
- Lead workshops, webinars, and presentations at scientific conferences on NGS advancements.

Key Skills: NGS expertise, troubleshooting, bioinformatics, customer support, and project management.

Shalamar Medical and Dental College, School of Allied Health – Lahore, Pakistan

Assistant Professor (Visiting)

[11/06/2024 – 15/07/2024]

- Teaching Under-Graduate and Post-Graduate Students (Bioinformatics and Genomics)

Biopharmaceutical and Biomarker Discovery lab, SBB, University of the Punjab, Lahore. Pakistan – Lahore, Pakistan

Research Associate

[18/07/2019 – 30/04/2022]

Development of a Dual-Property Anti-Inflammatory Biopharmaceutical for Rheumatoid Arthritis" project.

- Focused on evaluating immunomodulatory cytokines and exploring molecular mechanisms of inflammation.
- Contributed to the development of innovative therapeutic approaches for rheumatoid arthritis.

Applied advanced molecular techniques, including:

- qPCR - ELISA - SDS-PAGE - Protein purification - PCR-RFLPs - ARMS PCR - TaqMan assays
- Exome mapping, Proteomics, Big data analysis, STATA
- Conducted GWAS studies, genetic analysis, and meta-analyses to identify key biomarkers and therapeutic targets.
- Collaborated with teams and hospitals to develop treatments for rheumatoid arthritis.
- Engaged in clinical partnerships to advance the research.

 **Manchester Centre for Genomic Medicine** – Manchester, United Kingdom

Guest Researcher

[05/02/2019 – 13/06/2020]

 **National Centre of Excellence in Molecular Biology, University of the Punjab, Lahore, Pakistan** – Lahore, Pakistan

Research Associate

[06/06/2012 – 30/04/2014]

Involved in quality research with expertise in Biochemistry and Molecular Biology techniques.

- Proficient in:

- Bacterial culture Northern blotting - Southern blotting - Western blotting - DNA/RNA extraction - Plasmid DNA extraction
- RNAi technology - Transformation technology - Biosafety assessment studies of transgenic sugarcane

 **The Doctors lab** – Lahore, Pakistan

Biochemistry technologist

[20/08/2010 – 31/05/2012]

My involvement in this capacity was as follows:

- . I am well versed with the execution of Clinical biochemistry analysis (Liver function tests, Renal function tests, Electrolytes & minerals analysis, Miscellaneous)
- . Molecular diagnostic analysis (ELISA, PCR, RT PCR)
- . Sanger Sequencing

EDUCATION AND TRAINING

PhD Biochemistry

University of the Punjab [12/10/2015 – 09/08/2024]

City: Lahore | Country: Pakistan | Website: <https://www.pu.edu.pk/> | Field(s) of study: Biochemistry | Final grade: 3.22 | Thesis: Identification of Epilepsy Associated Molecular Signatures in Human Genomic Landscape using Whole Exome Sequencing

M.Phil Biochemistry

University of Veterinary and Animal Sciences, Lahore [10/10/2011 – 27/09/2013]

City: Lahore | Country: Pakistan | Website: <https://uvas.edu.pk/> | Field(s) of study: Biochemistry | Final grade: 3.54 | Thesis: Biosafety Studies of transgenic Sugarcane developed by CEMB

Bachelor Sciences (Hon`s) Biochemistry

The University of Lahore [12/10/2006 – 12/08/2010]

City: Lahore | Country: Pakistan | Website: <https://uol.edu.pk/> | Field(s) of study: Biochemistry | Final grade: 3.25 | Thesis: Apoptosis; A therapeutic Strategy in Eosinophilia

PUBLICATIONS

[2024]
Association of Epilepsy and Neurological Impairments with Homozygous Recessive Missense Mutations Found in the Genes Responsible for Ganglioside Biosynthesis (St3gal5) and Calcium Voltage-Gated Channels (Cacna1h)-Insights through Molecular Dynamic Simulations

Abid R, Nisar H, Chaudhary SU, Hamid M, Sahibzada KI, Firdous S, Mudassar M, Sadaf S

[2021]
Dysregulated Alanine as a Potential Predictive Marker of Glioma—an Insight from Untargeted Hrmass-Nmr and Machine Learning Data

Firdous, S, Rizwan A, Zubair N, Faisal Bukhari, Ammar Anwer, Leo L Cheng, and Saima S.

[2024]
Integrating HRMAS-NMR Data and Machine Learning-Assisted Profiling of Metabolite Fluxes to Classify Low- and High-Grade Gliomas

Safia Firdous, Zubair Nawaz, Rizwan Abid, Leo L. Cheng, Syed Ghulam Musharraf & Saima Sadaf

[2023]
A Novel Missense Compound Heterozygous Variant in Tlr1 Gene Is Associated with Susceptibility to Rheumatoid Arthritis—Structural Perspective and Functional Annotations

Usman Pasha · Kiran Hanif · Haseeb Nisar · Rizwan Abid · Muhammad U, M · Bilal Wajid · Saima Sadaf

[2023]
Molecular Dynamic Simulations Unravel the Underlying Impact of Missense Mutation in Autoimmunity Gene PTPN22 on Predisposition to Rheumatoid Arthritis

Pasha, Usman, Haseeb Nisar, Hajira Nisar, Rizwan Abid, Naeem Mahmood Ashraf, and Saima Sada

[2020]
Impact of IL-17f 7488t/C Functional Polymorphism on Progressive Rheumatoid Arthritis: Novel Insight from the Molecular Dynamic Simulations

[2021]
Analysis of Snp Rs1800796 Association with Risk of Rheumatoid Arthritis in Pakistani Population: A Case Control Study

Nisar, Haseeb, Usman Pasha, Kiran Hanif, Rizwan Abid, Rehan Ahmad Khan S, Haja N K, and Saima S.

[2021]
Analysis of Functional Polymorphisms in Meningioma associated Genes

Javaid, Anum, Binyamin Wattoo, Rizwan Abid, and Saima Sadaf.

[2013]
APOPTOSIS: A THERAPEUTIC STRATEGY IN EOSINOPHILIA

Muhammad Qasim1, Rizwan Abid, Mahmood-ur-Rahman, Muhammad S Masoud, Usman A Ashfaq

HIV OctaScanner: A Machine Learning Approach to Unveil Proteolytic Cleavage Dynamics in HIV-1 Protease Substrates". Journal of Chemical Information and Modeling

Sahibzada, Kashif; Shahid, Shumaila ; Akhter, Mohsina ; Abid, Rizwan; Azhar, Muteeba; Hu, Yuan-sen; Wei, Dong-Qing

[2025]

HCV genotyping and rational computational designing of an immunogenic multiepitope vaccine against genotype 3a Write here the description...

Authors: Kashif Iqbal Sahibzada · Rizwan Abid · Haseeb Nisar · Reham A. Rahman⁵ · Muhammad Idrees⁶ · Dong-Qing Wei^{7,8,9} · Yuansen Hu¹ · Saima Sadaf | **Volume, Issue and Pages:** 29, 2, 1-16 | **Publisher:** Springer Nature

Sahibzada, K. I., Abid, R., Nisar, H., Rahman, R. A., Idrees, M., Wei, D.-Q., Sadaf, S (2025). HCV genotyping and rational computational designing of an immunogenic multiepitope vaccine against genotype 3a. Journal of Computer Aided Molecular Design

PROJECTS

[10/06/2019 – 10/04/2022]

HEC-funded NRPU (National Research Program for Universities) Project title: Screening of Immunomodulatory Cytokines and Production of a Dual-Property Anti-Inflammatory Biopharmaceutical for Rheumatoid Arthritis. letter No. 001/8488/Punjab/NRPU/R&D/HEC/2017 dated 07-09-2018.(My role was as Research Associate)

[2012 – 2014]

Pakistan Agriculture Research Council Biosafety studies of transgenic sugarcane using animal models, CEMB/PARB/2014.(My role was as Research Associate)

CONFERENCES AND SEMINARS

[11/09/2022 – 17/09/2022]

Multi-omics Big Data Analyses Boot Camp

[18/10/2018 – 19/10/2018]

Royal Society of Chemistry, Pakistan 1st International conference on “medicinal chemistry and drug discovery” by COMSTECH, Islamabad

[23/10/2017 – 27/10/2017]

International Conference on “Trends and Prospects in Molecular Biosciences” by Institute of Biochemistry and Biotechnology, University of the Punjab, Lahore

[12/12/2016 – 13/12/2016]

Training Course on “Planning of cotton production, land preparation, weed control pest and disease control and harvesting” by CEMB, University of the Punjab

[25/08/2016 – 27/08/2016]

International Conference on “Recent Advances and Challenges in Molecular Biology, Biochemistry and Biotechnology” In CADR, Comsats, Abbottabad

[25/09/2013]

National Symposium on “Recombinant Enzymes for the Local Industry of Pakistan” in Institute of Biochemistry and Biotechnology, UVAS, Lahore

[29/11/2013 – 01/12/2013]

12th international shaukat khanum cancer symposium

[12/08/2025 – 12/08/2025]

The One-Day COMSTECH Hybrid Symposium titled as Preventing Inherited Disorders in Consanguineous Populations: A Framework for Action organized by OIC-Standing Committee on Scientific & Technological

NETWORKS AND MEMBERSHIPS

[17/09/2024 – Current]

Global Outreach Member (American Society of Microbiology)

SKILLS

Skills and Techniques acquired:

- Molecular techniques (DNA/RNA isolation, cloning, Primer/probe designing, PCR, RT- qPCR, TaqMan SNP Genotyping)
- Sanger Sequencing
- Next Generation (ILLUMINA)
- Microsoft Office(Excel, Powerpoint, Word)
- Statistical Tools (SPSS, GraphpadPrism, STATA)
- Bionformatics Softwares(Mega, CLC genomics,Geniuos Prime)
- Unix/Linux
- R programming language
- DRAGEN
- RNA Seq pipeline analysis, Variant Analysis

RECOMMENDATIONS

Name: Prof. Dr. Muhammad Iqbal | Director ORIC, University of Lahore

- I have know them for several years...

Email: iqbalm@fulbrightmail.org | Phone number: (+92) 3234318393

Name: Prof. Dr. Idrees Nasir | Professor

prof. Idrees Nasir was my M.Phil. supervisor.

Email: dr.idrees@gmail.com | Phone number: (+92) 3217440016

Name: Dr. Safee Ullaah Chaudhary | Associate Professor

I have worked with Dr. Safee Ullah on a project.

Email: safee.ullah.chaudhary.2@gmail.com | Phone number: (+92) 3214255171